



1585 - Analysis of GTO Observations of Uranus by the MIRI Instrument

Cycle: 1, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Observations of Uranus by the MIRI instrument made in the GTO program 1248 will be analyzed for spatial variability of temperatures, composition and possible stratospheric hazes. We concentrate this effort on three particular topics. (1) Determine the deep tropospheric structure. Its global mean is a key to understanding the accuracy of a model whose spectrum is now used as a calibration standard for submillimeter and longer wavelengths. This is very important in light of our ongoing analysis on Herschel and ISO observations that imply a protosolar mixture of helium and hydrogen, making Uranus unlike the helium-depleted atmospheres of giant planets Jupiter and Saturn. Testing this model critically involves examining the center-to-limb structure of 9-11 micron radiances to diagnose the presence of a warm, emitting particulate layer in the stratosphere. (2) Determine upper-tropospheric/lower- stratospheric temperatures and spin states of H₂, providing an indirect constraint on vertical motions. (3) Determine the temperature structure and H₂ spin state of the stratosphere using center-to-limb behavior of H₂ quadrupole and dimer lines that will distinguish between alternative models derived from analysis of Spitzer IRS observations. These results will be compared with Voyager-2 IRIS, Spitzer IRS (disk-averaged), and spatially resolved ground-based thermal images and spectra. This comparison will clarify ambiguities in their analysis and measure long-term evolution of atmospheric properties one season after the 1986 spatially resolved Voyager-2 measurements of thermal emission.

OBSERVING DESCRIPTION

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The analysis will address archived observations by the MIRI experiment made in the GTO program 1248, for which Heidi Hammel is the PI and which has been implemented by Leigh Fletcher.

The observations consist of three sets of measurements by all 12 settings of the MIRI instrument using medium-resolution spectroscopy in IFU mode with four-point dithering to ensure coverage of the whole disk. Each set of measurements will be made at approximately 120 degrees of longitude separation in order to cover the planet at all longitudes during its 17h 14m rotation. On the advice of the MIRI team, one observation of the background sky will also be taken.

Program 1248 also includes NIRSPEC observations, which will not be addressed by this program except to supply context in reflected sunlight should a major atmospheric feature in the MIRI spectral images be detected.